

OFFICIAL PRODUCT BROCHURE

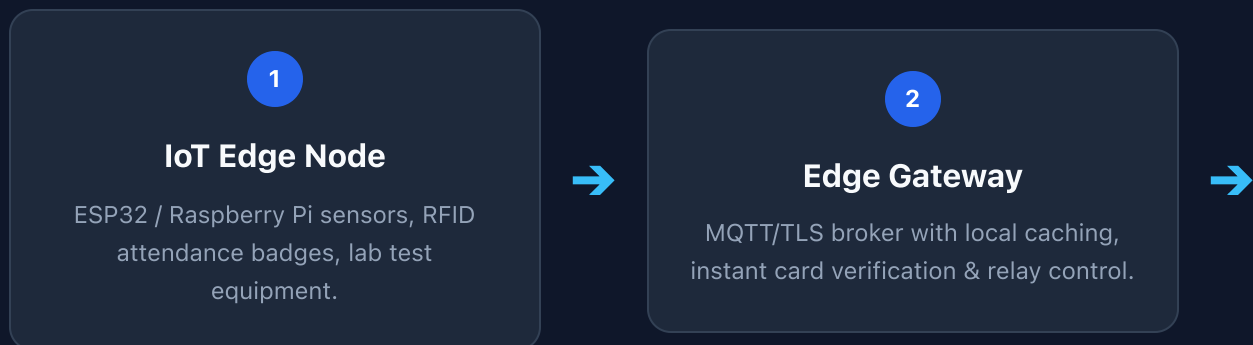
# Matrix-IoT Platform

Hardware-to-Cloud Education & Lab Automation System hosted at matrix-iot.com and integrated with Matrix LMS Cloud.

## End-to-End System Architecture

Bridging physical microcontrollers, edge gateways, real-time sensor telemetry, and automated college grading.

### System Data Flow & Topology



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**Matrix Cloud LMS**

Supabase Row-Level Security, multi-tenant college isolation & REST API.



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**Real-time Analytics**

Student attendance log, auto-graded lab assessments, and device health monitoring.

## Key Capabilities

Built specifically for technical colleges, engineering universities, and smart campuses.

**Automated RFID Lab Attendance**

Students swipe their smart badges at the lab entry. Attendance records automatically sync to Matrix LMS with millisecond latency and tamper-proof verification.

**Hardware-Based Lab Grading**

Connect microcontrollers directly to assignment test benches. Code uploads are validated on physical target devices with real-time feedback to student dashboards.

**Multi-Tenant Cloud Security**

Tenant isolation guaranteed via Supabase Row Level Security (RLS). Every college's telemetry and student records remain encrypted and segregated.

**MQTT & WebSockets Engine**

Supports low-latency streaming of sensor data (temperature, voltage, logic states) with fallback offline buffer for network drops.



MicroPython & C++ SDKs

Includes plug-and-play libraries for ESP32, STM32, Arduino, and Raspberry Pi with pre-configured API authentication wrappers.



Campus Telemetry Dashboard

Real-time analytics for lab managers detailing active devices, energy usage, student lab session durations, and hardware fault alerts.

Technical Specifications

Hardware compatibility, cloud protocols, and security standards.

| Category               | Supported Specification / Standard                                                        |
|------------------------|-------------------------------------------------------------------------------------------|
| Hardware Compatibility | ESP32 Dual-Core 240MHz, Raspberry Pi 4/5/Zero, STM32, Arduino Uno/Mega, PN532 RFID        |
| Networking Protocols   | MQTT v3.1.1 / v5 over TLS 1.3, HTTPS REST API, WebSockets, CoAP                           |
| Database & Security    | Supabase PostgreSQL, AES-256 Cloud Storage, JWT Bearer Token Auth, Row Level Security     |
| Sensor Integration     | DHT22/DHT11, Analog Voltage Probes, Digital Logic Analyzers, Relay Modules, OLED Displays |
| Latency & Reliability  | <120ms cloud sync time, local EEPROM failover buffer up to 10,000 offline scans           |

# Deploy Matrix-IoT in Your Campus Today

Host your smart campus labs at **[matrix-iot.com](https://matrix-iot.com)** or seamlessly integrate into Matrix LMS Cloud.

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